

22-10-2024 25K 7 PL/SQL Procedures, functions, Loops and Procedure for loops

AIM:

To implement PL/SQL procedure function and loops on number theory and business scenarios

1. Simple PL/SQL program (static input)

DECLARE

message VARCHAR2(20) = 'Booking closed';

BEGIN

dbms_output.put_line(message);

END;

Output:

Booking closed

2. Conditional Statement (Dynamic Input)

DECLARE

hid NUMBER(2) = 100;

BEGIN

if (hid = 10) THEN

dbms_output.put_line('Value of hid is 10');

ELSE IF (hid = 20) THEN

dbms_output.put_line('Value of hid is 20');

ELSE IF (hid = 30) THEN

dbms_output.put_line('Value of hid is 30');

ELSE IF (hid = 30)

dbms_output.put_line('None of the value's matching');

END IF;

dbms_output.put_line('Exact value of hid is (hid)');

END;

Output:

None of the value is matching

Exact value of hid 10 is 100

3. Nested loops Example

DECLARE

hid NUMBER(1);

oid NUMBER(1);

BEGIN

<<outer-loop>>

FOR hid IN 1..3 LOOP

<<inner-loops>>

FOR oid IN 1..3 LOOP

dbms_output.put_line

END LOOP inner-loop;

END LOOP outer-loop;

END;

Output:

hid is:1 and oid is:1

hid is:1 and oid is:2

hid is:1 and oid is:3

hid is:2 and oid is:1

hid is:2 and oid is:2

hid is:2 and oid is:3

hid is:3 and oid is:1

hid is:3 and oid is:2

hid is:3 and oid is:3

4. Procedure Example

CREATE OR REPLACE PROCEDURE booking_status

<cid IN NUMBER>

IS

BEGIN

IF cid > 200 THEN

dbms_output.put_line('No booking available');

ELSE

dbms_output.put_line('Booking open');

END IF

END;

Execution

BEGIN

booking-status (150);

booking-status (250);

END;

Output

Booking open

~~No Booking available~~

FOR PL/SQL PROCEDURE FOR LOOPS

Example 1: Using WHILE loop with cursor

CREATE OR REPLACE PROCEDURE print_prime_customerID;

CURSOR cust_cur IS

SELECT customer_id FROM customers;

V_id NUMBER;

V_is_prime BOOLEAN;

V_i NUMBER;

BEGIN

OPEN cust_cur;

LOOP

FETCH cust_cur INTO V_id;

EXIT WHEN cust_cur

OF V_id < 2 THEN

V_is_prime := FALSE;

ELSE

V_is_prime := TRUE;

V_i := 2;

WHILE V_i <= TRUNC(SQRT(V_id)) LOOP

IF MOD(V_id, V_i) = 0 THEN

V_is_prime := FALSE;

EXIT;

END IF;

V_i := V_i + 1;

END LOOP;

END IF;

IF V_is_prime THEN

DBMS_OUTPUT.PUT_LINE('Prime customer ID: ' || V_id);

END IF;

END LOOP;

CLOSE cust_cur;

END;

The procedure checks are customer IDs in the table and prints the prime ones using a while loop

Example 2: Using for loop for first N prime numbers

CREATE OR REPLACE PROCEDURE print-first-n-primes
(n number) IS

V-num NUMBER := 2

V-count NUMBER := 0;

V-is-prime BOOLEAN;

BEGIN

WHILE V-count < n loop

V-is-prime := True;

FOR i IN 2 TRUNC(SORT(V-num)) LOOP

IF MOD(V-num) = 0 THEN

V-is-prime := FALSE;

EXIT;

ENDIF;

END LOOP;

IF V-is-prime THEN

DBMS_OUTPUT.PUT_LINE ('Prime: ' || V-num);

V-count := V-count + 1

ENDIF;

V-num := V-num + 1

END LOOP;

END;

The procedure prints the first n prime number using a for loop.

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RECORD (5)	5
(20)	5

RESULT:

Thus the procedure function and loops program using PL/SQL procedures, functions and loops are executed successfully.

